

Subject Description Form

Subject Code	COMP5121
Subject Title	Data Mining and Data Warehousing Applications
Credit Value	3
Level	5
Pre-requisite/ Exclusion	Nil
Objectives	The objectives of this subject are to enable students to: 1. make more effective use of data stored in databases; 2. create a clean, consistent repository of data within a data warehouse; 3. utilize various levels and types of summarization of data to support management decision making; and 4. discover patterns and knowledge that is embedded in the data using different data mining techniques.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) acquire deep understanding on the need for data warehouse in real applications; b) master the typical data warehouse architectures, and be able to analyze and identify the components of data warehouse architecture, addressing complex interdisciplinary issues with professionalism; c) be able to apply and design a data warehouse in support of business problem solving in real applications for societal benefit and sustainability; d) develop sophisticated insights on typical knowledge discovery process and the different algorithms available by popular commercial data mining software, leveraging emerging technologies for global challenges; and e) obtain hands-on experience with some popular data mining software.
Subject Synopsis/ Indicative Syllabus	• Introduction to data warehousing and data mining; possible application areas in business and finance; definitions and terminologies; types of data mining problems. • Data warehouse and data warehousing; data warehouse and the industry; definitions; operational databases vs. data warehouses. • Data warehouse architecture and design; two-tier and three-tier architecture; star schema and snowflake schema; data characteristics; static and dynamic data; meta-data; data marts. • Data replication, data capturing and indexing, data transformation and cleansing; replicated data and derived data; Online Analytical Processing (OLAP); multidimensional databases; data cube. • Data Mining and knowledge discovery, the data mining lifecycle; pre-processing; data transformation; types of problems and applications.

	• Mining of Association Rules; the A priori algorithm; binary, quantitative and generalized association rules; interestingness measures. • Classification; decision tree based algorithms; Bayesian approach; statistical approaches, nearest neighbor approach; neural network based approach; Genetic Algorithms based technique; evaluation of classification model. • Clustering; k-means algorithm; Hierarchical algorithm; Condorset; neural network and Genetic Algorithms based approach; evaluation of effectiveness. • Sequential data mining; time dependent data and temporal data; time series analysis; sub-sequence matching; classification and clustering of temporal data; prediction. • Computational intelligence techniques; fuzzy logic, genetic algorithms and neural networks for data mining.																																	
Teaching/Learning Methodology	Class activities including - lecture, tutorial, lab, workshop seminar where applicable A mix of lectures, discussions and case study analysis.																																	
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific Assessment Methods/Tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>Assignments, Tests & Projects</td><td>30</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Final Examination</td><td>70</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Total</td><td>100</td><td></td><td></td><td></td><td></td><td></td></tr></table>	Specific Assessment Methods/Tasks	% weighting	Intended subject learning outcomes to be assessed					a	b	c	d	e	Assignments, Tests & Projects	30	✓	✓	✓	✓	✓	Final Examination	70	✓	✓	✓	✓		Total	100					
Specific Assessment Methods/Tasks	% weighting			Intended subject learning outcomes to be assessed																														
		a	b	c	d	e																												
Assignments, Tests & Projects	30	✓	✓	✓	✓	✓																												
Final Examination	70	✓	✓	✓	✓																													
Total	100																																	
Student study effort expected	<table><tr><td colspan="2">Class Contact:</td></tr><tr><td>Class activities (lecture, tutorial, lab)</td><td>39 hours</td></tr><tr><td colspan="2">Other student study effort:</td></tr><tr><td>Assignments, Quizzes, Projects, Exams</td><td>66 hours</td></tr><tr><td>Total student study effort</td><td>105 hours</td></tr></table>	Class Contact:		Class activities (lecture, tutorial, lab)	39 hours	Other student study effort:		Assignments, Quizzes, Projects, Exams	66 hours	Total student study effort	105 hours																							
Class Contact:																																		
Class activities (lecture, tutorial, lab)	39 hours																																	
Other student study effort:																																		
Assignments, Quizzes, Projects, Exams	66 hours																																	
Total student study effort	105 hours																																	
Reading list and references	(1) Han, J., Pei, J., and Tong, H., 2022, Data Mining: Concepts and Techniques, 4th Ed., Morgan Kaufmann. (2) Tan, PN., Steinbach, M., Karpne, A., and Kumar, V., 2018, Introduction to Data Mining (Second Edition): https://www-users.cse.umn.edu/~kumar001/dmbook/index.php (3) Foulds, J., Witten, I. H., Frank, E., Hall, M. A., and Pal, C. J., 2025, Data Mining: Practical machine learning tools and techniques. Elsevier. (4) Shmueli, G., Bruce, P. C., Yahav, I., Patel, N. R., and Lichtendahl Jr, K. C, 2017, Data mining for business analytics: concepts, techniques, and applications in R. John Wiley & Sons. (5) Maimon, O. Z., and Rokach, L., 2014, Data mining with decision trees: theory and applications (Vol. 81). World scientific. (6) Schutt, R., and O'Neil, C, 2014, Doing data science: Straight talk from the frontline (pp. 181-182). O'Reilly.																																	

	(7) Kimball, R., and Ross, M, 2013, The data warehouse toolkit: The definitive guide to dimensional modeling. John Wiley & Sons. (8) ACM SIGKDD Conferences on Knowledge Discovery and Data Mining: https://kdd.org/conferences (9) ACM CIKM Conferences on Information and Knowledge Management: https://dl.acm.org/conference/cikm/proceedings (10) IEEE International Conferences on Data Mining (ICDM): https://ieeexplore.ieee.org/xpl/conhome/1000179/all-proceedings (11) UC Irvine Machine Learning Repository of Databases: https://cml.ics.uci.edu/ and https://archive.ics.uci.edu/
--	--